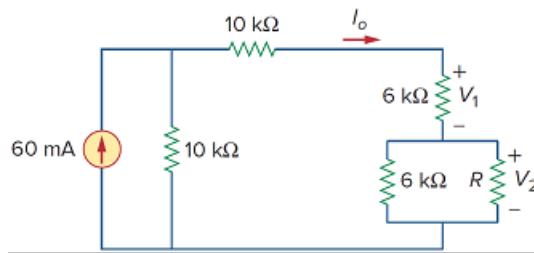


Problem

In the circuit of Fig. 3.102, find the values of R , V_1 , and V_2 given that $i_o = 15 \text{ mA}$.

Figure 3.102:



Step-by-step solution

Step 1 of 4

Refer to circuit diagram in Figure 3.102 in the text book.

Identify the loops and current s in the circuit diagram shown in Figure 1.

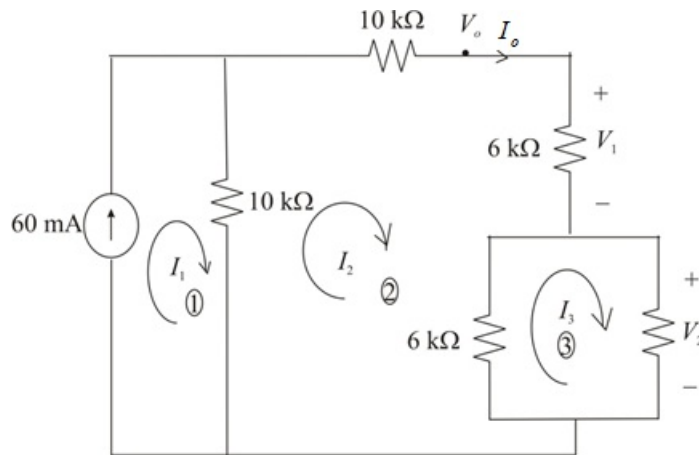


Figure 1

Step 2 of 4

The value of the current i_o is 15 mA . The value of the current I_2 is equal to i_o . That is,

$$I_2 = 15 \text{ mA}$$

Calculate the value of the voltage V_1 .

$$V_1 = I_2 (6 \text{ k})$$

Substitute 15 mA for I_2 .

$$\begin{aligned} V_1 &= (15 \text{ m})(6 \text{ k}) \\ &= 90 \text{ V} \end{aligned}$$

Therefore, the value of the voltage V_1 is, $\boxed{90 \text{ V}}$.

Step 3 of 4

Calculate the value of the voltage V_o .

$$V_o = (I_1 - I_2)(10 \text{ k}) - I_2(10 \text{ k})$$

Substitute 15 mA for I_2 and 60 mA for I_1 .

$$\begin{aligned} V_o &= (60 \text{ m} - 15 \text{ m})(10 \text{ k}) - (15 \text{ m})(10 \text{ k}) \\ &= (45 \text{ m})(10 \text{ k}) - (15 \text{ m})(10 \text{ k}) \\ &= 450 - 150 \\ &= 300 \text{ V} \end{aligned}$$

Calculate the value of the voltage V_2 .

$$V_2 = V_o - V_1$$

Substitute 300 V for V_o and 90 V for V_1 .

$$\begin{aligned} V_2 &= 300 - 90 \\ &= 210 \text{ V} \end{aligned}$$

Therefore, the value of the voltage V_2 is, 210 V.

Step 4 of 4

Consider the expression for the voltage V_2 .

$$V_2 = (I_3 - I_2)(6 \text{ k})$$

Substitute 15 mA for I_2 and 210 V for V_2 .

$$210 = (I_3 - 15 \text{ m})(6 \text{ k})$$

$$I_3 - 15 \text{ m} = \frac{210}{6 \text{ k}}$$

$$I_3 - 15 \text{ m} = 35 \text{ m}$$

$$I_3 = 50 \text{ mA}$$

Calculate the value of the resistance R .

$$R = \frac{V_2}{I_3}$$

Substitute 50 mA for I_3 and 210 V for V_2 .

$$\begin{aligned} R &= \frac{210}{50 \text{ m}} \\ &= 4.2 \text{ k}\Omega \end{aligned}$$

Therefore, the value of the resistance R is 4.2 kΩ.